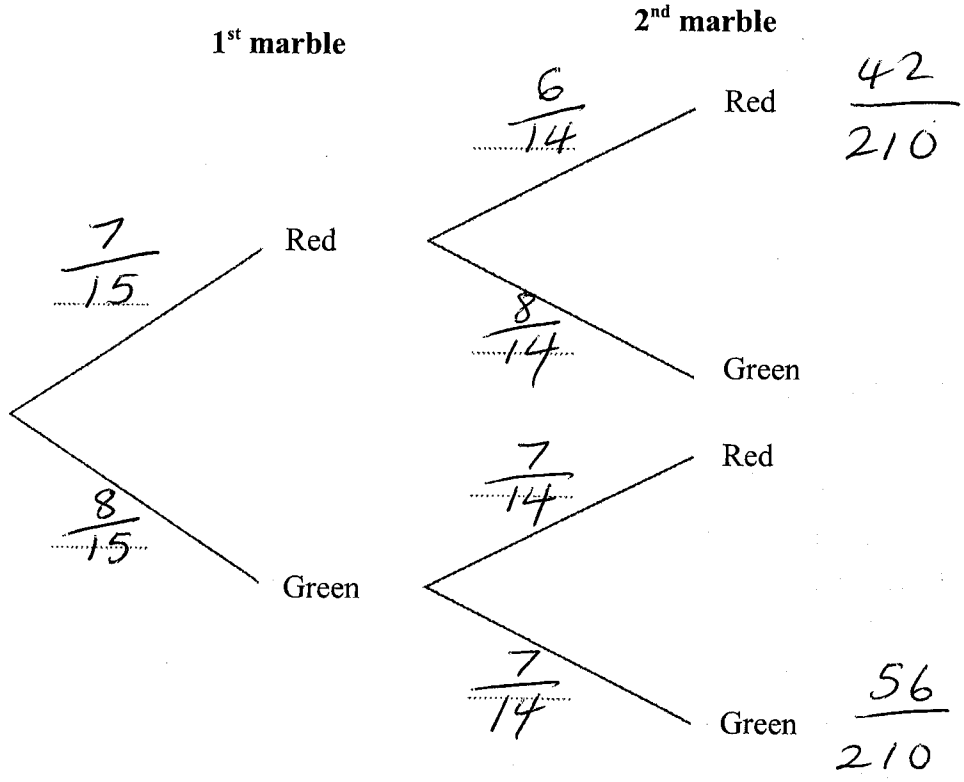
1 There are only red marbles and green marbles in a bag.
There are 7 red marbles and 8 green marbles.

Mason takes at random a marble from the bag.
He does not put the marble back in the bag.

Mason takes at random a second marble from the bag.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Mason takes two marbles the same colour.

$\frac{7}{15} \times \frac{6}{14} = \frac{42}{210}$

$\frac{8}{15} \times \frac{7}{14} = \frac{56}{210}$

$\frac{56}{210}$

(2)

(Total for Question 1 is 4 marks)

(OR $\frac{7}{15}$)

2 By completing the square, find the coordinates of the turning point of the curve with the equation $y = x^2 + 8x - 3$
You must show all your working.

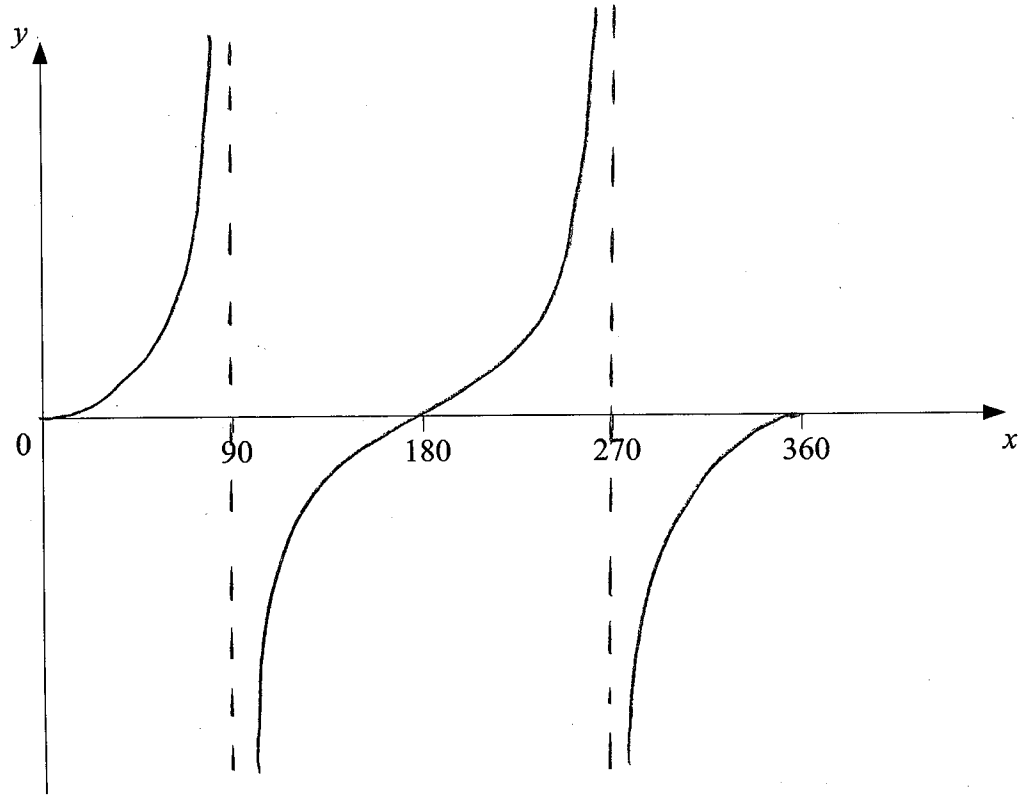
$(x+4)^2 - 16 - 3$

$(x+4)^2 - 19$

$(-4, -19)$

(Total for Question 2 is 3 marks)

3 Sketch the graph of $y = \tan x°$ for $0 \leq x \leq 360$



(Total for Question 3 is 2 marks)

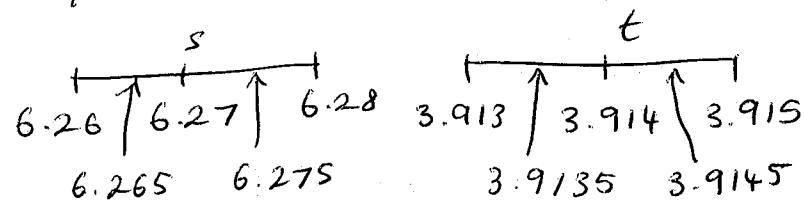
4

$$v = \frac{s}{t}$$

s = 6.27 correct to 2 decimal places
t = 3.914 correct to 3 decimal places

Work out the upper bound for v.

Give your answer to 3 decimal places.



$$\text{upper } v = \frac{\text{upper } s}{\text{lower } t}$$

$$= \frac{6.275}{3.9135}$$

$$= 1.603$$

1.603

(Total for Question 4 is 3 marks)

---

5 Solve $\frac{4}{x-2} + \frac{2}{x-5} = 3$

$$4(x-5) + 2(x-2) = 3(x-2)(x-5)$$

$$4x - 20 + 2x - 4 = 3(x^2 - 5x - 2x + 10)$$

$$6x - 24 = 3(x^2 - 7x + 10)$$

$$6x - 24 = 3x^2 - 21x + 30$$

$$0 = 3x^2 - 27x + 54$$

$$0 = x^2 - 9x + 18$$

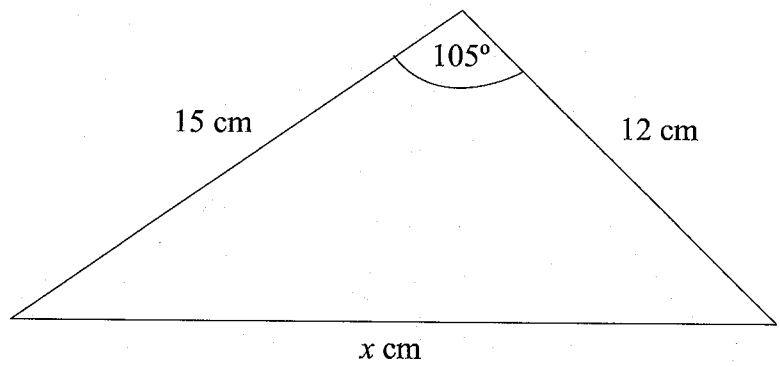
$$(x - 6)(x - 3) = 0$$

$x = 6$ $\quad$ $x = 3$

$x = 3$ or $x = 6$

(Total for Question 5 is 4 marks)

6



Work out the value of x.
Give your answer to 1 decimal place.

$$a^2 = b^2 + c^2 - 2bc \cos A$$
$$= 15^2 + 12^2 - 2(15)(12)\cos(105)$$
$$= 462.17\ldots$$
$$a = \sqrt{462.17}$$
$$= \underline{\underline{21.5}}$$

21.5

(Total for Question 6 is 3 marks)

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7 Prove that the difference between the squares of any 2 consecutive integers is equal to the sum of these integers.

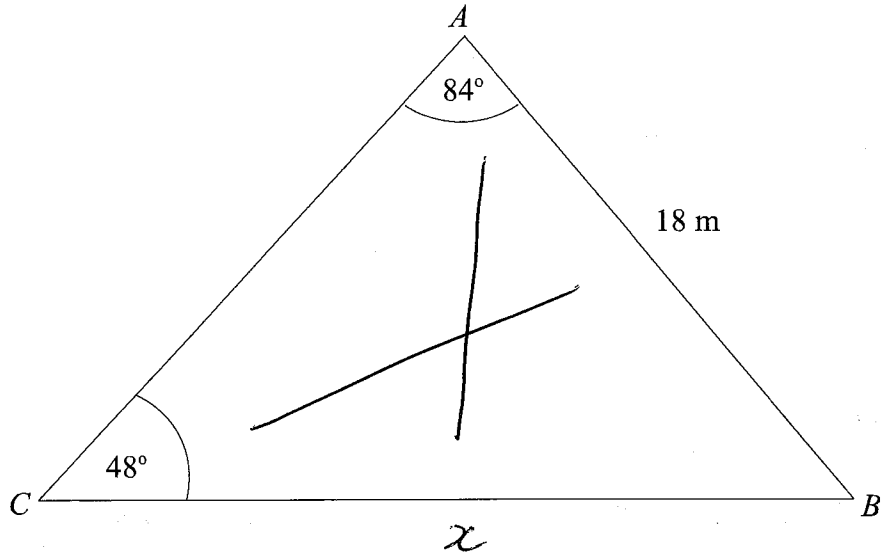
$$(n+1)^2 - n^2$$
$$(n+1)(n+1) - n^2$$
$$n^2 + n + n + 1 - n^2$$
$$\underline{\underline{2n + 1}}$$

$$n + (n+1)$$
$$\underline{\underline{2n + 1}}$$

$$2n + 1 = 2n + 1$$

(Total for Question 7 is 3 marks)

8



Work out the area of triangle *ABC*

Give your answer to 1 decimal place.

$$\frac{x}{\sin 84} = \frac{18}{\sin 48}$$

$$x = \frac{18}{\sin 48} \times \sin 84$$

$$= 24.0887 \text{ m}$$

Angle ABC $= 180 - 48 - 84 = \underline{48^\circ}$

$$\text{Area} = \frac{1}{2}(18)(24.0887)\sin 48$$

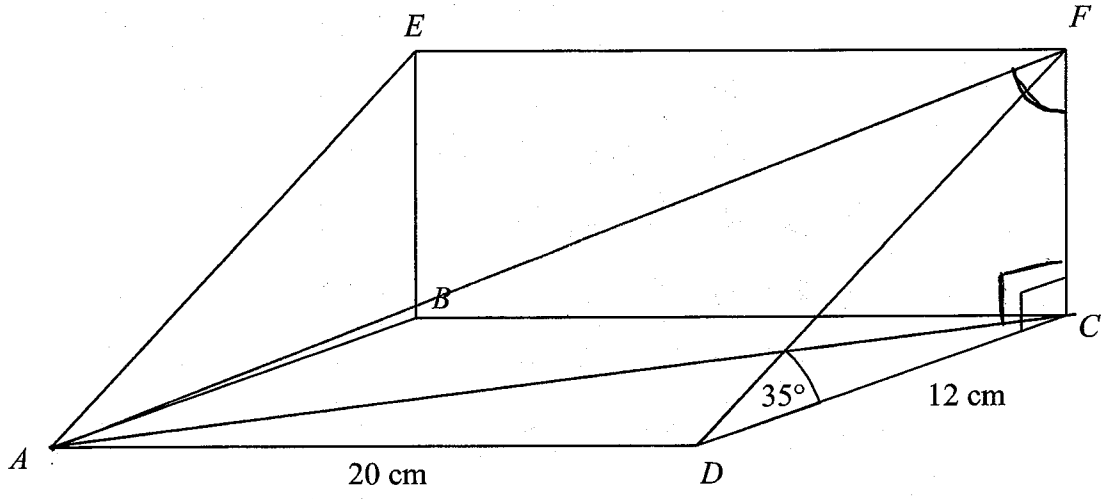
$$= \underline{161.1 \text{ m}^2}$$

161.1 ..... m^2

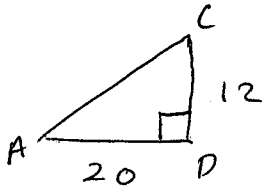
(Total for Question 8 is 5 marks)

9 The diagram shows a triangular prism.

$CD = 12$ cm
$AD = 20$ cm
Angle $ADC = 35°$



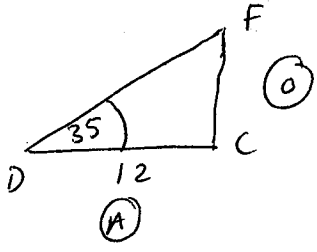
Calculate the size of angle AFC.
Give your answer correct to 1 decimal place.



$AC^2 = 12^2 + 20^2$

$AC = \sqrt{12^2 + 20^2}$

$= 4\sqrt{34}$

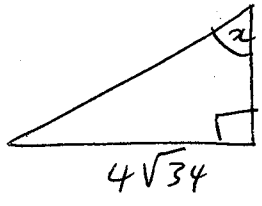


O

$\tan 35 = \frac{CF}{12}$

$CF = 12 \tan 35$

$= 8.402$



8.402 A

O

$\tan x = \frac{4\sqrt{34}}{8.402}$

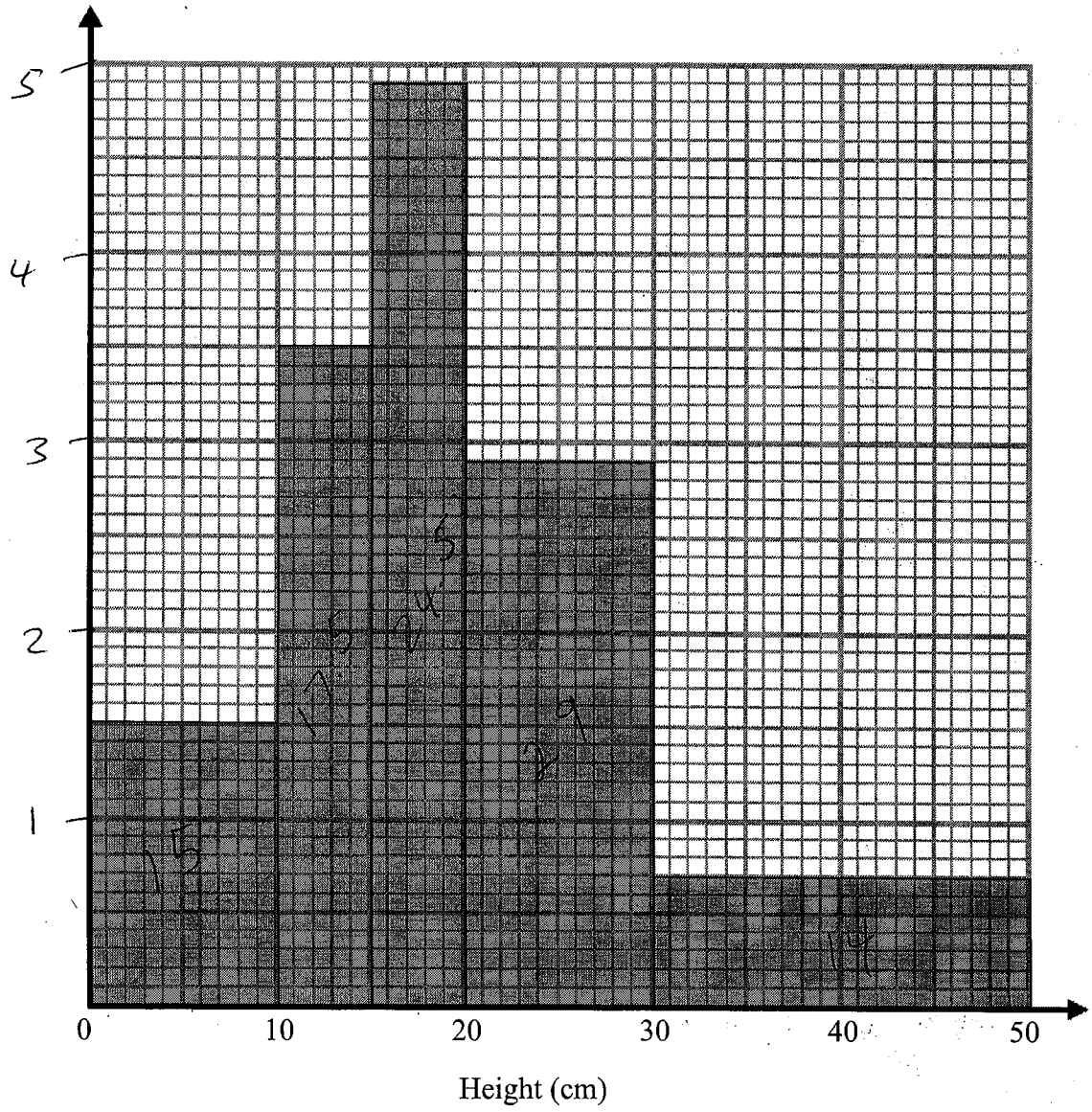
$x = \tan^{-1}(2.7758)$

$= 70.2°$

70.2 °

(Total for Question 9 is 4 marks)

10 The histogram shows information about the height of some plants.



(a) Work out an estimate for the proportion of plants over 25cm tall.

$15 + 17.5 + 24.5 + 29 + 14 = 100$

$\frac{29}{2} + 14 = 28.5$ $\quad \frac{28.5}{100} = \frac{57}{200}$

$\frac{57}{200}$

(3)

(b) Explain why your answer to part a is only an estimate.

we do not know the exact heights of the plants we assumed the plants in the 20-30 cm group were evenly distributed.

(1)

(Total for Question 10 is 4 marks)